

Intervals and absolute value equations

Polynomial-size vertex test for inverse M-matrix intervals

Difficulty: challenging **Importance:** interesting to specialist**Status:** Open

Rating rationale: Replacing an exponential interval test by a quadratic vertex family is challenging; verified inverse-positivity has specialist importance in interval matrix analysis.

Area: interval linear algebra; structured matrices

Problem statement

A real square matrix M is a nonsingular M-matrix if its off-diagonal entries are nonpositive, M is invertible, and M^{-1} is entrywise nonnegative. An inverse M-matrix is the inverse of such a matrix.

For every $n \geq 1$, let $L, U \in \mathbb{R}^{n \times n}$ satisfy $L \leq U$ entrywise, and set

$$\mathcal{A} = \{A : L \leq A \leq U\}, \quad C = (L + U)/2, \quad R = (U - L)/2.$$

For $i = 1, \dots, n$, let $z^{(i)}$ have entry -1 in position i and entry 1 elsewhere, and write $D_i = \text{diag}(z^{(i)})$.

Is the following equivalence true?

$$\begin{aligned} &[\text{every } A \in \mathcal{A} \text{ is an inverse M-matrix}] \\ &\iff \\ &\left[\begin{array}{l} C + sD_iRD_j \text{ is an inverse M-matrix} \\ \text{for every } i, j \in \{1, \dots, n\} \text{ and } s \in \{-1, 1\} \end{array} \right]. \end{aligned}$$

All entries in the interval vary independently, and degenerate intervals are allowed. The proposed test uses at most $2n^2$ matrices; repeated test matrices are harmless. This would give a small exact certificate for a structured property needed in interval inversion and verified linear-system computation.

References

Milan Hladík, *An overview of polynomially computable characteristics of special interval matrices*, in *Beyond Traditional Probabilistic Data Processing Techniques*, Springer (2020), pp. 295–310; [author preprint](#), [arXiv:1711.08732v1](#), §9, Conjecture 1, p. 11, with the vectors defined in Theorem 24.

Jürgen Garloff, Doaa Al-Saafin, and Mohammad Adm, *Further Matrix Classes Possessing the Interval Property*, *Reliable Computing* **28** (2021), 56–70, p. 64, paragraph immediately before IP 4.4.

Status check

The 2021 paper explicitly retains this conjecture while proving a different sufficient vertex family of size 2^{n-1} . On 2026-09-10, searches combining the original title, inverse M-matrices, interval property, conjecture, and 2025/2026 found no resolution. Hladík's current publication list and the later interval-property paper were checked. This is a selected-source status check, not an exhaustive citation audit.

Independent audit — 2026-09-10

The full Hladík preprint, §9, Conjecture 1, agrees with the displayed two-sign vertex family, including the definition of each $z^{(i)}$. Garloff–Al-Saafin–Adm, p. 64, expressly leaves that conjecture unresolved; its exponential family is a different test, not a resolution for a subclass here. Independent later searches for the exact conjecture and matrix class found no resolution.